Module 6 Basic Content:

Following are specific job categories that can be found within and related to the water/wastewater industry. This is not an exhaustive list, but many different job categories are represented – and most people do not realize they are related to the water/wastewater industry. Note that for larger systems, there may be one or many people who fill a specific job category but in smaller systems, there may be one person whose job description includes multiple job categories. For example, a small rural water system of less than 500 customers may have a single water operator who is also the lab tech, heavy equipment operator, office manager, bookkeeper, and communications person; they do not fulfill each of these roles on a daily basis but likely do so on a monthly or bi-monthly basis. In this situation, you just hope that there is not a major line break in the middle of sending out water bills.

- Water Operators/Technicians: In this career, water techs' predominant duties are to ensure that water is piped from the source (surface or groundwater), treated at the water treatment facility as required to ensure it is clean and free from disease-causing organisms, stored appropriately, and piped out to households on the water system. Water techs are responsible for both the infrastructure and the water carried within it, with the division of responsibility for the two about even. As a career it combines working inside with working outside and seldom are two days are alike. Water techs need a variety of skills to address the many contexts in which they work; including working on pumps, collecting water samples, keeping maintenance logs, using heavy equipment, laying water line, using chemicals to treat water to remove pathogens and toxins, and working with the public, just to name a few. Water techs are often one of only one or two employees for the water system, so they need to be self-driven, organized, critical thinkers, and problem-solvers to manage the many planned tasks and unexpected situations that arise each day. To be a water tech, you need a high school diploma or GED, specialized training in water treatment, and must pass required state licensure exams.

- Wastewater Operators/Technicians: Wastewater techs hold similar duties to water techs, but with the added responsibility of handling and treating wastewater and biologics. In this career, wastewater techs are predominantly responsible for ensuring the wastewater that enters the sewer system makes it to the wastewater treatment facility, is properly treated, and returned to the environment in a manner that has the least impact on the environment. Wastewater techs are responsible for both the infrastructure and the wastewater carried by it, with the division of responsibility for the two weighted more in favor of the wastewater treatment. Wastewater techs need

a variety of skills because they have the potential to work in a variety of situations, but most of their time will be spent at the water treatment plant. The process of separating water from the waste and biologics found in it is a multi-step process requiring several phases of chemical/UV treatment, filtration, and aeration. Due to this process, wastewater techs need a stronger grasp of basic science and math to perform the many tests and treatments correctly. Wastewater techs also need to be critical thinkers and problem-solvers who can think on their feet. To be a wastewater tech, you need a high school diploma or GED, specialized training in wastewater treatment, and must pass required state licensure exams.

- Onsite Wastewater System Operators/Technicians: This career focuses on site evaluation, set up, and management of decentralized wastewater systems. In many places these are simply home septic systems, but decentralized systems can be much more complex, including small-scale multi-step systems similar to those found in larger wastewater treatment plants, but which serve smaller communities. Onsite wastewater techs share similar duties with water/wastewater techs, so they must be self-driven, organized, critical thinkers, and problem-solvers who can think on their feet. Onsite techs also have the added responsibility of site evaluation to determine the type and placement of new systems, which requires specialized knowledge related to soils and subsurface transport processes. To be an onsite wastewater system tech, you need a high school diploma or GED, specialized training in wastewater treatment, and must pass required state-specific licensure exams.

- Laboratory Scientists/Technicians: This career focuses on water quality and tends to be housed in a water lab but may also include going out in the field to collect samples. Lab techs perform various tests on water and wastewater samples to determine the presence or absence of biologics or chemical substances. The range of tests can be extensive, so it is important that lab techs have a strong understanding of science - especially biology and chemistry. Lab scientists go beyond testing to investigation or experimentation and often manage the lab. Lab techs/scientists must be organized, detail-oriented, inquisitive, critical thinkers, and problem-solvers. To be a lab tech you need at minimum a bachelor’s degree; to be a lab scientist you often need master’s degree or PhD.

- Heavy Equipment Operators: This career focuses on building or maintaining infrastructure. In the water and wastewater industry, heavy equipment operators most often dig trenches for waterlines, lateral lines, and sewer systems or remove the earth to get to a what needs to be repaired or moved,

but they will be involved in anything that requires moving large amounts of soil around. Heavy equipment operators need to be reliable, safety-conscious, spatially oriented, critical thinkers, and problem-solvers. To be a heavy equipment operator, you need a high school diploma or GED, and specialized training in operating various pieces of heavy equipment. A CDL is not required to operate heavy equipment, but it is recommended as many heavy equipment operators personally transport their own equipment for business purposes.

- Office Managers: This career focuses on the business end of the water/wastewater industry and deals with the public. In most water/wastewater systems, the office manager is the person housed in the main office who may have many different responsibilities – depending on the size of the system. This person may manage other employees, answer the phone and route that information to the correct employee, handle customer billing (accounts receivable), pays bills for the system (accounts payable), research potential expenditures for the system, communicate with outside businesses, integrate/implement/maintain software (billing, telemetry), maintain records for employees and board members (licensing, training), and communicate with and plan/execute board meetings of the local system managing board. While this is a long list, it does not capture everything office managers may be responsible for. Office managers need to be outgoing, organized, planful, good at communicating, have a strong grasp of basic accounting methods, be tech savvy, good at conflict resolution, and enjoy new challenges. Office managers in small systems often do the work of many of the career categories that follow. To be an office manager, you need a high school diploma or GED, but an associate's or bachelor's degree in management with coursework in accounting/bookkeeping is highly recommended.

- Accountants/Bookkeepers: This career focuses on maintaining the accounts/ books for the system – from small water systems to large municipalities. This means they keep track of the payments that come in, the payments that go out, money moved from one account to another, account analysis, and making recommendations based on future projections from that analysis. This person may also be responsible for managing tax information for the system. Bookkeepers tend to manage the "accounts ledger" while accountants tend to be more involved with analysis and planning. Accountants/bookkeepers need to be trustworthy, organized, and have a strong working knowledge of ethical accounting practices. A college degree is not necessarily required to be a

bookkeeper, but it is recommended. An accountant must have at minimum a bachelor's degree and must pass a certification exam to be a licensed CPA.

- Information Technology Specialist: This career focuses on managing the technology-based aspects of a water/wastewater system and is more likely to be found in larger wastewater systems and municipalities since those systems have many more "moving parts". In most cases, drinking water treatment requires little intervention (filtration and disinfection) so it is easier to maintain oversight on the process doing it the old-fashioned way. Centralized wastewater treatment always requires a multi-step process with wastewater moving from treatment tank to treatment tank across a huge facility. Because of this, many larger systems use computer-based systems to monitor and manage the multiple unit operations within the plant. The computer-based systems are used to monitor the water/wastewater flowing through the plant as well as the infrastructure and equipment involved in treatment (tank volume, pumps, filters, agitators, sluice gates, valves, line pressure, chemical dispensers, etc.) The IT specialist's job is to maintain/update/replace the software/hardware that is used to oversee the operation of the system, be able to query the software to pinpoint problems, and may also be called on to manage drone technology, which is the latest technology used to monitor areas that need to be physically viewed but are physically difficult to access. IT specialists are also expected to keep abreast of the most recent updates or changes in technology/software and provide technology-related recommendations upon request. IT specialists need to be lifelong learners to keep up with the continuous changes in the IT industry. They also need to be organized, planful, forward-thinking, situationally aware, and be good at math or computer programming. A college degree is not necessarily required to be an IT specialist, so it is important to note that many career tech programs offer certificates in different aspects of information technology. If seeking promotion, a bachelor's degree is recommended. Seeking out continuing education credits (CEUs) on an annual basis is also recommended to keep current.

- Media/Communication Specialist: This career focuses on communication with and outreach to the community and is more likely to be found in a large municipal system. Media/communication specialists provide the "public face" for the system by maintaining/managing the website, reaching out to the community using various forms of social media, and possibly producing ad campaigns (for example, when a bond is being floated that raises funding for the municipal system, it is important to convince the community to support the bond). Media/communication specialists need to be good communicators,

organized, planful, forward-thinking, situationally aware, and creative. A college degree is not necessarily required to be media/communication specialist, so it is important to note that many career tech programs offer certificates in different aspects of digital media production. If seeking promotion, a bachelor's degree is recommended. Seeking out continuing education credits (CEUs) on an annual basis is also recommended to keep current.

- Civil Engineers: This career focuses on designing, planning, and building large infrastructure; large wastewater systems and municipalities are likely to have a civil engineer on staff, but smaller systems are likely to contract work with a civil engineer who focus on either infrastructure or structural development – or both – when needed. Civil engineers are the people who figure out how to get water to and wastewater from customers when proposals for new housing developments or new industry are put on a municipality's drawing board. They are also the ones who figure out the best way to extract water from new drinking water sources, deliver it to the water system, and clean the water (design the water treatment plant) prior to sending it to customers. Civil engineers must see the big picture and also make plans to bring the big picture to reality; they are organized, creative, forward-thinking, planful, problem-solvers, communicators, who are very good at math. Civil engineering requires a minimum of a bachelor's degree from an accredited program, licensure is required to provide services publicly, and a master's degree is commonly required for promotion.

- Environmental Engineers: An environmental engineer is a specific type of civil engineer with a different focus. This career focuses on the place where humans and the environment intersect and works to limit the effect humans have on the environment. Large wastewater systems and municipalities are likely to have an environmental engineer on staff, but smaller systems are likely to contract work with an environmental engineer when needed. In good circumstances, environmental engineers are the people who *figure out how to prevent* environmental contamination from human outputs (sewage, heavy metals, chemicals, smoke, etc.). In bad circumstances, they must *quickly figure out how to clean up* environmental contamination from human catastrophes (spills of many forms: crude oil, raw sewage, automobile fuel, nuclear waste, rocket fuel, etc.). The phrase "an ounce of prevention is worth a pound of cure" might be a common motto for these folks. Environmental engineers must see the big picture and how all parties involved will be affected by the decisions made from their recommendations; they are critical thinkers, problem-solvers, organized, forward-thinking, planful,

communicators, who are very good at math and science – particularly biology and soil science. Environmental engineering requires a minimum of a bachelor's degree and to be licensed the bachelor's degree must be from an accredited program. A minimum of a master's degree is commonly required for promotion.

- Global Positioning System (GPS) Field Technician: This career focuses on collecting GPS data and processing it for import into a GIS map. GPS field technicians often use GPS data collection units which are handheld or mounted on heavy equipment to gather GPS data points. Each point of data is collected using the GPS unit as it triangulates information from orbiting GPS satellites. The GPS unit records the locational information from the GPS satellites and this data is stored in the unit for later download. After downloading, the data is post-processed, verified for accuracy, and exported as GIS files which are then imported into a geodatabase. This data will be used later to create maps and web apps. GPS field technicians need to be detail-oriented, reliable, good communicators, and enjoy working outdoors. To be a GPS field technician you need a high school diploma or GED and specialized training in GPS data collection. Many career techs/colleges/universities offer certificate programs which provide the specialized training for this career.

- Geographic Information System (GIS) Specialists/Analysists: This career focuses on using GPS data and specialized software to visually represent all forms of data through the creation of maps and web apps for querying said maps. GIS maps are often interactive and hold much more information than a regular street map. Consider the map app on your phone. It not only provides you with street names and cardinal directions, like a paper map does, it also provides you with business names, their websites, hours of operation, menus, customer reviews, directions for how to get there, and much more. For water/wastewater systems, those maps might include water line sizes, type and location of pumps and valves, location of fire hydrants and water towers, date of last inspection, install date, and many other things. All that data is based on geographic information systems. Each data point on the map can have many different types of information layered in it, depending on the data sets used by the GIS specialist/analysist. They often work closely with GPS field technicians and endpoint users to ensure that their product is correct and meets the needs of stakeholders. GIS specialist/analysist need to be detail-oriented, spatially oriented, life-long learners (everchanging software), analytical, problem-solvers, and tech-savvy. To be a GIS specialist/analysist, a bachelor's degree is recommended *but not necessarily in GIS*. Many

current GIS specialist/analysist in the field have a science degree (environmental science, geology, biology) and a GIS certificate. A GIS specialist/analysist specializes in using a particular software, so this training has often been embedded in a different degree or added on as a certificate or continuing education credits. Only recently have degrees in geospatial technology come into the scene. Many career techs/colleges/universities offer certificate programs in GIS which are readily accepted in industry.

- Source Water Protection Specialists: Surface Water Protection Specialist and Groundwater Protection Specialist are two other names for this career. This career focuses on preventing the contamination of sources of drinking water – whether surface water, groundwater, or both. Source water protection specialists inventory all known potential contaminant sources (industry, small businesses like hair salons, gas stations, or junk yards, farms and ranches/feedlots, wastewater treatment plants, personal septic systems, etc.) within the watershed of a drinking water source and produce a protection plan for the water system(s) which use that water source. Protection plans are predominantly educational in nature and work to create voluntary partnerships to ensure that affected stakeholders are aware of potential contaminants and take precautionary steps to prevent those contaminants from entering the water source. Source water protection specialists need to have a strong science background (something related to environmental science or hydrology), be detail-oriented, critical thinkers, problem-solvers, effective communicators, able to use GPS/GIS (source water protection plans include GIS maps), and since they work with the public so much, they especially need to be outgoing flexible, and diplomatic. Source water protection specialists require a minimum of a bachelor’s degree, and for promotion, a minimum of a master’s degree is commonly required.